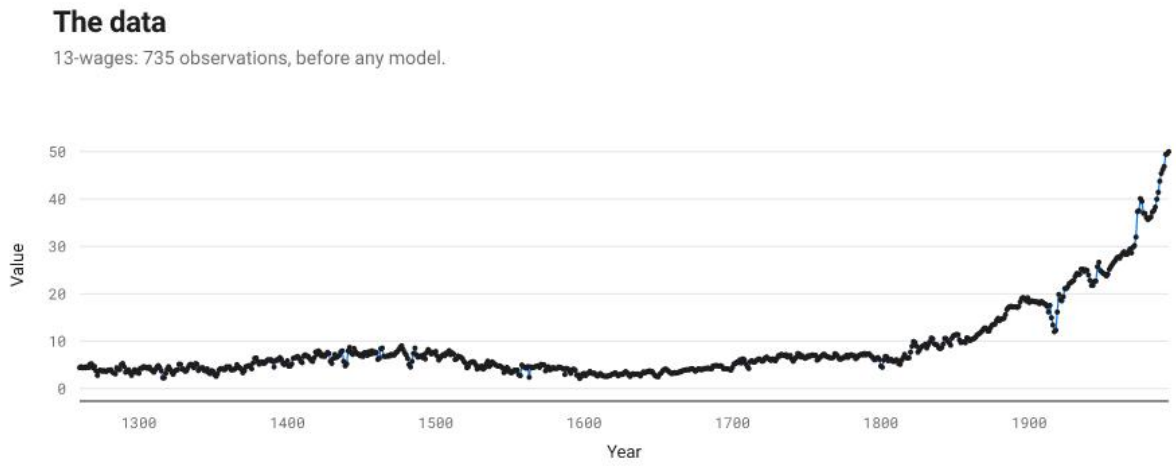
An automatic report for the dataset: 13-wages

Some Math empowered by some AI guided by some researchers

09 November 2026

Produced by `autostat`*, a Python 3 port of the Automatic Bayesian Covariance Discovery system of Lloyd et al. (AAAI 2014).*



The data

Executive summary

The structure search selected `CW(WN, C) + Lin x Lin x SE + Lin x SE + Lin x Lin x SE + Lin x Lin x SE` (Bayesian information criterion 1350.2; negative log likelihood 609.1 on 735 points). It decomposes into 6 additive components, listed by how much each reduces extrapolation error:

1. A smooth function with linearly increasing marginal standard deviation.
2. A constant. This function applies from 1604 until 1704.
3. A rapidly varying smooth function with quadraticly varying marginal standard deviation.
4. A rapidly varying smooth function with quadraticly varying marginal standard deviation.
5. Uncorrelated noise. This function applies until 1604 and from 1704 onwards.
6. A smooth function with quadraticly varying marginal standard deviation.

#	R^2 cumulative (%)	Residual variance explained (%)	CV MAE	MAE reduction (%)
data			8.76	
1	96.6	96.6	1.59	81.9
2	96.6	0.0	1.58	0.7
3	97.2	18.2	1.57	0.2
4	98.4	40.6	1.57	0.1
5	98.4	2.2	1.57	0.0
6	100.0	100.0	1.6	-1.5

MAE: mean absolute error over ten contiguous held-out blocks; CV: cross-validated.

The formula found

The structure search is a symbolic regression: it searches a grammar of expressions over kernels and keeps the one with the lowest Bayesian information criterion. This is the expression it kept, with every fitted parameter in place; t is in years.

$$y(t) = f_1(t) + f_2(t) + f_3(t) + f_4(t) + f_5(t) + f_6(t)$$

Each component is a Gaussian process with its own covariance:

$$f_1(t) \sim \mathrm{GP}\left(0,\ \ 0.001103\,(t-1127)(t'-1127)\cdot\exp\left(-\frac{(t-t')^2}{2\cdot 148.3^2}\right)\right)$$

$$f_2(t) \sim \mathrm{GP}\big(0,\ \ w_1(t)\,w_1(t')\,[0.05652]\big)$$

$$f_3(t) \sim \mathrm{GP}\left(0,\ \ 2.48\times 10^{-11}\,(t-1074)(t'-1074)\cdot(t-1776)(t'-1776)\cdot\exp\left(-\frac{(t-t')^2}{2\cdot 0.97^2}\right)\right)$$

$$f_4(t) \sim \mathrm{GP}\left(0,\ \ 4.588\times 10^{-11}\,(t-1194)(t'-1194)\cdot(t-1633)(t'-1633)\cdot\exp\left(-\frac{(t-t')^2}{2\cdot 1.883^2}\right)\right)$$

$$f_5(t) \sim \mathrm{GP}\big(0,\ \ (1-w_1(t))(1-w_1(t'))\,[0.06744\,\delta_{tt'}]\big)$$

$$f_6(t) \sim \mathrm{GP}\left(0,\ \ 1.475\times 10^{-10}\,(t-1204)(t'-1204)\cdot(t-1672)(t'-1672)\cdot\exp\left(-\frac{(t-t')^2}{2\cdot 14.95^2}\right)\right)$$

The changes of regime weigh the components with:

$$w_1(t) = \frac{1}{4}\left(1+\tanh\left(2344\,(t-1605)\right)\right)\left(1-\tanh\left(2344\,(t-1705)\right)\right)$$

These components have a closed form, read from their posterior mean:

$$f_2(t) = -0.2425\,w_1(t)$$

The other components are smooth, quasi-periodic or noise without a short formula: a draw from such a process has no finite closed form; its covariance above is the formula.

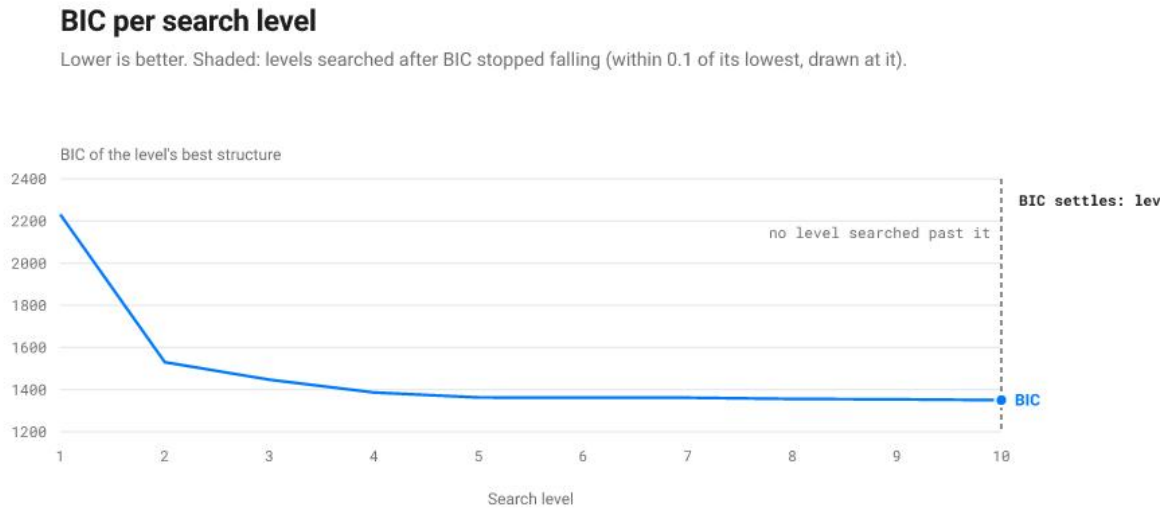
Closed form of each component:

#	component	closed form
1	A smooth function with linearly increasing marginal standard deviation	none: the covariance is the formula
2	A constant. This function applies from 1604 until 1704	exact polynomial of degree 0
3	A rapidly varying smooth function with quadraticly varying marginal standard deviation	none: the covariance is the formula
4	A rapidly varying smooth function with quadraticly varying marginal standard deviation	none: the covariance is the formula
5	Uncorrelated noise. This function applies until 1604 and from 1704 onwards	none: the covariance is the formula
6	A smooth function with quadraticly varying marginal standard deviation	none: the covariance is the formula

Search path and stopping criterion

The 2014 configuration searches a fixed number of levels and keeps the lowest BIC; it has no stopping rule. The table follows the winner of each level. A change of structure adds or removes terms; a level that keeps the structure only refits its parameters. BIC differences read on the scale of Kass and Raftery (1995): under 2 barely worth mentioning, 2 to 6 positive, 6 to 10 strong, over 10 very strong.

level	BIC	change	structure	components	closed	params
1	2231.7		WN + SE	2	0	3
2	1529.9	-701.8	WN + Lin x WN + Lin x SE + SE	4	0	8
3	1447.2	-82.6	WN + Lin x Lin x WN + Lin x Lin x SE + Lin x SE	4	0	11
4	1386.1	-61.1	WN + Lin x Lin x WN + Lin x Lin x SE + Lin x Lin x SE + Lin x SE	5	0	15
5	1362.8	-23.4	WN + Lin x Lin x SE + Lin x Lin x SE + Lin x SE	4	0	12
6	1361.6	-1.1	WN + Lin x Lin x SE + Lin x Lin x SE + Lin x SE	4	0	12
7	1361.6	+0.0	WN + Lin x Lin x SE + Lin x Lin x SE + Lin x SE	4	0	12
8	1355.7	-5.9	CW(WN, C) + Lin x Lin x SE + Lin x SE + Lin x Lin x SE + Lin x Lin x SE	6	1	20
9	1354.1	-1.7	CW(WN, C) + Lin x Lin x SE + Lin x SE + Lin x Lin x SE + Lin x Lin x SE	6	1	20
10 (kept)	1350.2	-3.8	CW(WN, C) + Lin x Lin x SE + Lin x SE + Lin x Lin x SE + Lin x Lin x SE	6	1	20



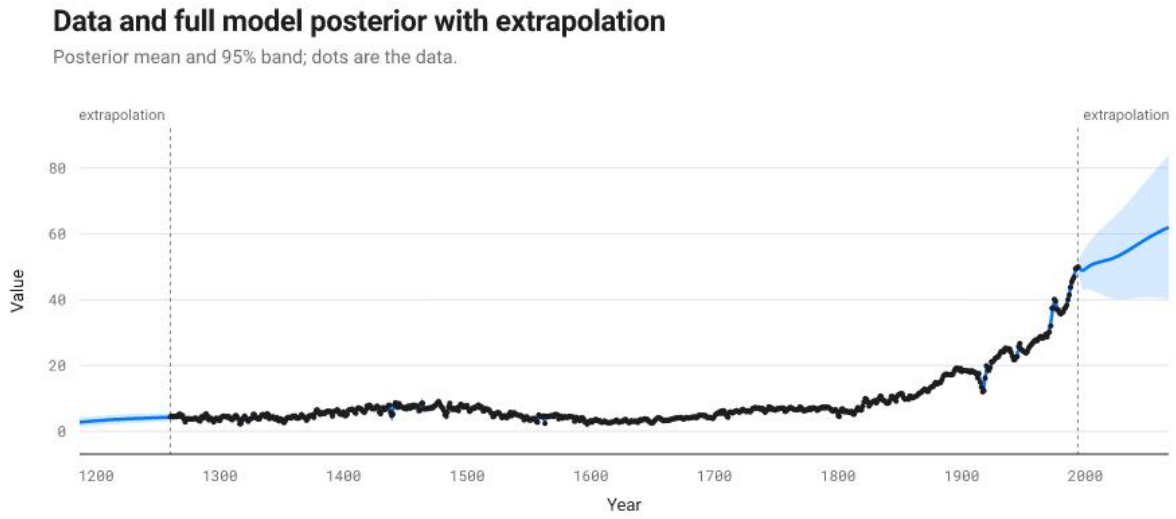
BIC per search level

Posterior weight of each structure met, exp(-BIC/2) normalised under a flat prior, after the Bayesian machine scientist (Guimerà et al., 2020): where the search keeps one structure, the weights say how much the data prefer it to the others. The 4 structure(s) under 0.001 are left out.

structure	lowest BIC	weight
CW(WN, C) + Lin x Lin x SE + Lin x SE + Lin x Lin x SE + Lin x Lin x SE	1350.2	0.997
WN + Lin x Lin x SE + Lin x Lin x SE + Lin x SE	1361.6	0.003

- The search ran 10 levels (the published 2014 search: ten levels, no stopping rule) and keeps the model of level 10, CW(WN, C) + Lin x Lin x SE + Lin x SE + Lin x Lin x SE + Lin x Lin x SE, BIC 1350.2.
- That structure was first reached at level 8; the levels after it only refitted its parameters, lowering BIC by 5.5 in all.
- The best different structure, WN + Lin x Lin x SE + Lin x Lin x SE + Lin x SE (level 6), is 11.4 above it: very strong evidence for the kept one on the Kass and Raftery scale.
- Weighed by exp(-BIC/2) under a flat prior, as the Bayesian machine scientist weighs expressions, the structures met give the kept one a posterior weight of 0.997 and the next, WN + Lin x Lin x SE + Lin x Lin x SE + Lin x SE, 0.003: the kept one takes almost all the weight.
- The last level lowered BIC by 3.8 through its parameters only: the structure had settled, its fit had not quite.
- A rule stopping after 1 level(s) without a fall above 0.1 would have ended the search at level 7 and kept WN + Lin x Lin x SE + Lin x Lin x SE + Lin x SE (level 6, BIC 11.4 above).
- A rule stopping after 2 level(s) without a fall above 0.1 would not have stopped within these 10 levels.

- The kept expression has 20 parameters for 735 points: its BIC is a fit term 2 NLL = 1218.2 plus a complexity term p ln n = 132.0, each extra parameter costing 6.60; it splits into 6 components, 1 of them with a closed form.



Data and full posterior

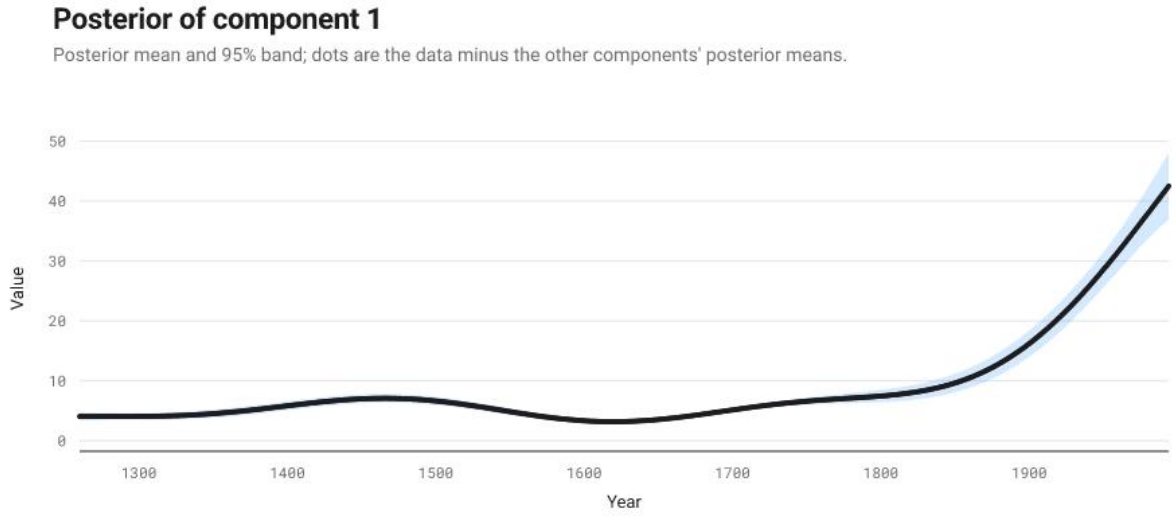
Detailed discussion of the components

Component 1: A smooth function with linearly increasing marginal standard deviation

This component is a smooth function with a typical lengthscale of 148.3 years. The marginal standard deviation of the function increases linearly.

This component explains 96.6% of the data's variance on its own; with the components above it, 96.6%. Adding it changes the cross-validated MAE by -81.9%.

Extrapolation. This component is assumed to continue smoothly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on smooth functions with a marginal mean of zero and a typical lengthscale of 148.3 years. The marginal standard deviation of the function is assumed to continue to increase linearly.



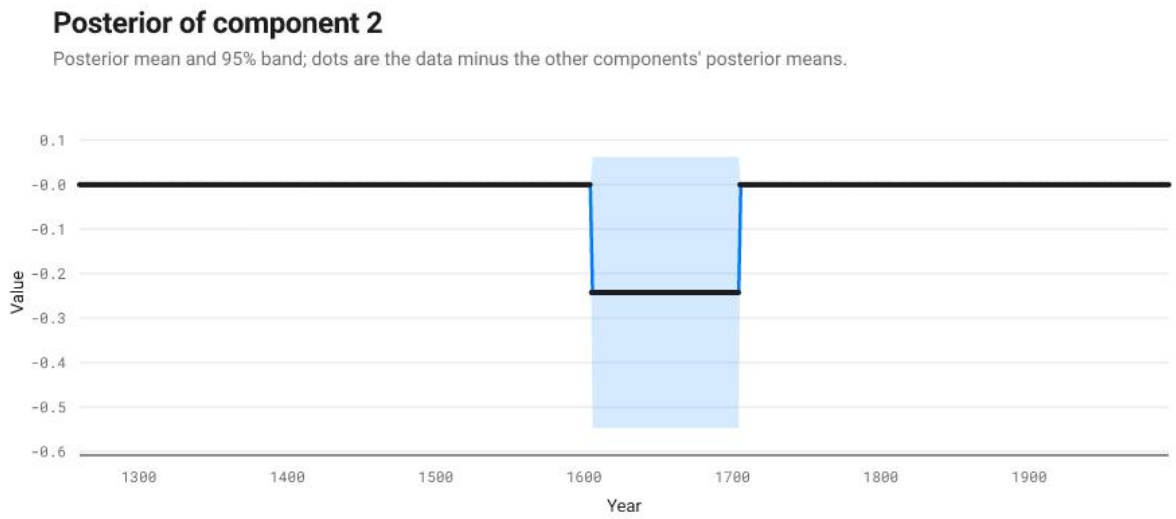
Component 1

Component 2: A constant. This function applies from 1604 until 1704

This component is constant. This component applies from 1604 until 1704.

This component explains 0.5% of the data's variance on its own; with the components above it, 96.6%. Adding it changes the cross-validated MAE by -0.7%.

Extrapolation. This component is assumed to stop before the end of the data and will therefore be extrapolated as zero.



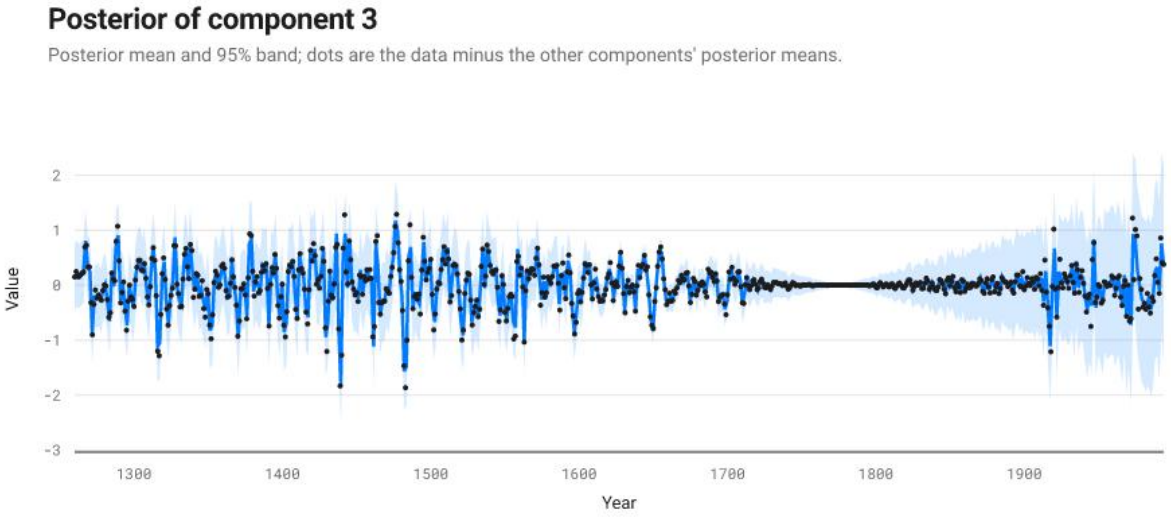
Component 2

Component 3: A rapidly varying smooth function with quadraticly varying marginal standard deviation

This component is a rapidly varying but smooth function with a typical lengthscale of 1.0 years. The marginal standard deviation of the function varies quadraticly.

This component explains 0.6% of the data's variance on its own; with the components above it, 97.2%. Adding it changes the cross-validated MAE by -0.2%.

Extrapolation. This component is assumed to continue smoothly but its distribution is assumed to quickly return to the prior. The prior distribution places mass on smooth functions with a marginal mean of zero and a typical lengthscale of 1.0 years. The marginal standard deviation of the function is assumed to continue to vary quadraticly.



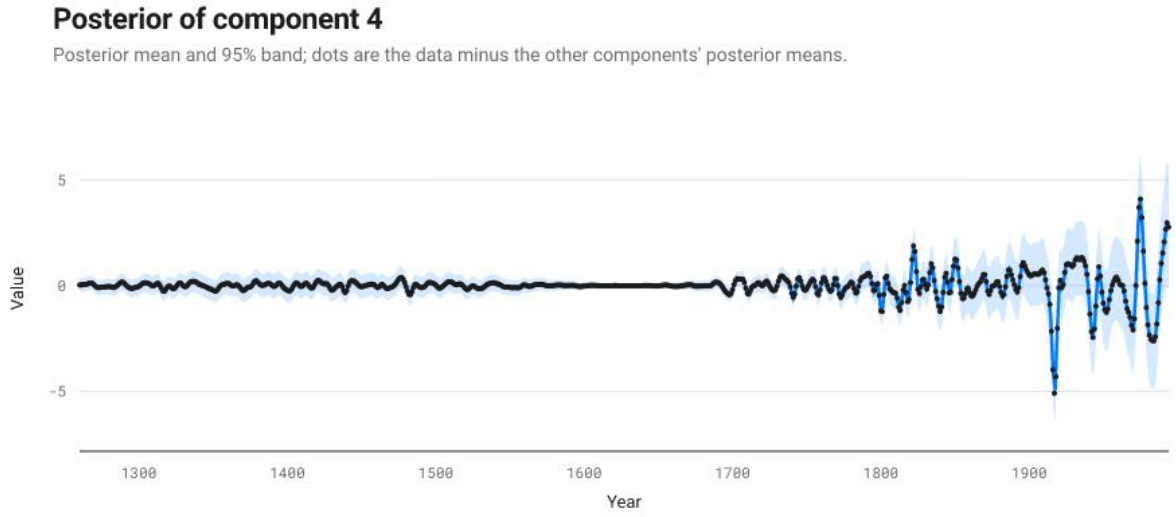
Component 3

Component 4: A rapidly varying smooth function with quadraticly varying marginal standard deviation

This component is a rapidly varying but smooth function with a typical lengthscale of 1.9 years. The marginal standard deviation of the function varies quadraticly.

This component explains 1.2% of the data's variance on its own; with the components above it, 98.4%. Adding it changes the cross-validated MAE by -0.1%.

Extrapolation. This component is assumed to continue smoothly but its distribution is assumed to quickly return to the prior. The prior distribution places mass on smooth functions with a marginal mean of zero and a typical lengthscale of 1.9 years. The marginal standard deviation of the function is assumed to continue to vary quadraticly.



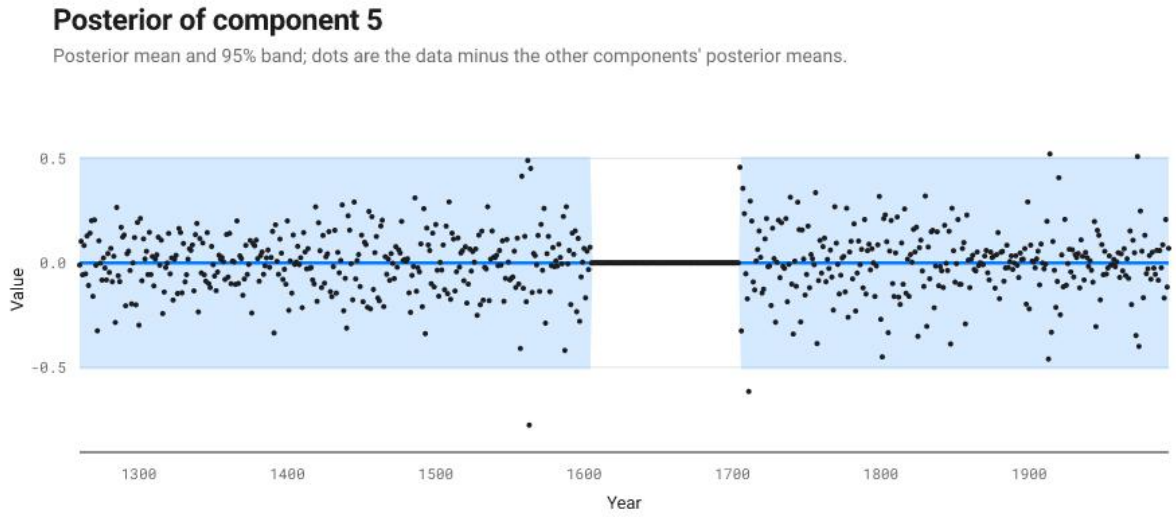
Component 4

Component 5: Uncorrelated noise. This function applies until 1604 and from 1704 onwards

This component models uncorrelated noise. This component applies until 1604 and from 1704 onwards.

This component explains 0.1% of the data's variance on its own; with the components above it, 98.4%. Adding it changes the cross-validated MAE by -0.0%.

Extrapolation. This component assumes the uncorrelated noise will continue indefinitely.



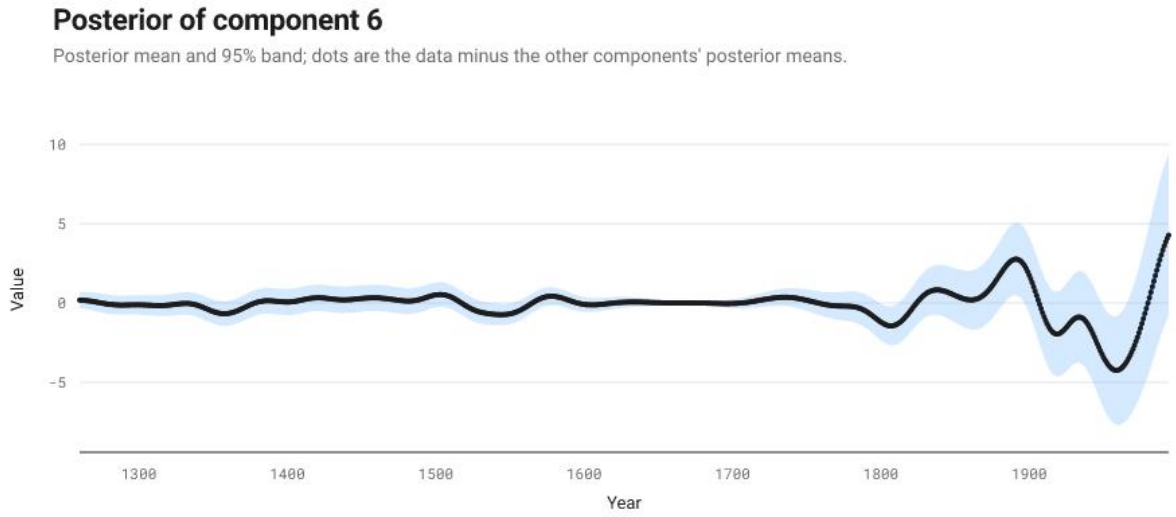
Component 5

Component 6: A smooth function with quadraticly varying marginal standard deviation

This component is a smooth function with a typical lengthscale of 14.9 years. The marginal standard deviation of the function varies quadraticly.

This component explains -4.2% of the data's variance on its own; with the components above it, 100.0%. Adding it changes the cross-validated MAE by 1.5%.

Extrapolation. This component is assumed to continue smoothly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on smooth functions with a marginal mean of zero and a typical lengthscale of 14.9 years. The marginal standard deviation of the function is assumed to continue to vary quadraticly.



Component 6

Model checking

Each statistic compares draws of a component's posterior with draws of its prior. A small p-value (bold below 0.05) points at a property of the data that the component's prior does not reproduce.

#	QQ max	QQ min	ACF min	ACF min lag	Pgram max	Pgram peak	MMD
1	0.074	0.206	**0.004**	0.054	0.490	0.256	**0.000**
2	0.414	0.558	0.990	0.996	0.740	0.960	**0.034**
3	0.854	0.654	0.954	0.926	0.944	0.944	**0.000**
4	0.820	0.420	0.960	0.800	0.916	0.986	**0.000**
5	0.930	0.676	0.980	0.966	0.984	0.994	0.095
6	0.626	0.630	0.854	0.858	0.800	0.990	**0.000**

QQ: quantile-quantile gaps. ACF: autocorrelation function. Pgram: periodogram. MMD: maximum mean discrepancy two-sample test.